Chapter 1 Homotopy Groups

Recall:

- Component π_0;
- Fundamental Group π_1;

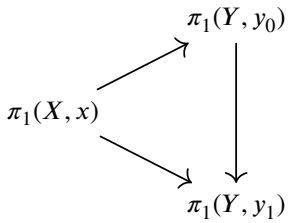
And with covering $p : X \to Y$, we obtain the induced injection

$$\pi_1(p, x) : \pi_1(X, x) \to \pi_1(Y, y)$$

Especially, for a G-covering, an exact sequence

$$1 \to \pi_1(X, x) \overset{\pi_1(p,x)}{\to} \pi_1(Y, y) \overset{\partial}{\to} G \to 1$$

;
Seifert-van Kampen:
$Aut_{\prod(X)}(x)$;
Homotopy $f_0, f_1 : X \to Y$, then we obtain a group isomorphism $\pi_1(Y, y_0) \cong \pi_1(Y, y_1)$:



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Consider the following triple (X, A, x). The **long exact sequence**

$$\pi_1(A, x) \overset{\pi_1(i,x)}{\to}$$

1.1 Homotopy Group

Definition 1.1.0.1 Define:

- I^n; $\partial I^n = \{(s_1, \dots, s_n) | \exists i, s_i \in \{0, 1\}\}$
- Homotopy Classes $\pi_n(X, x) = \{[f] | f : (I^n, \partial I^n) \to (X, \{x\})\}$;

Then: Group Structure

- Multiplication:

$$f * g := \begin{cases} f(2s_1, s_2, \dots, s_n) \\ f(2s_1 - 1, s_2, \dots, s_n) \end{cases}$$

- Identity:$s \mapsto x$;
- Inverse: $f^-(s_1, \dots, s_n) := (1 - s_1, s_2, \dots, s_n)$;

♠ **Proposition 1.1.0.2** The group $\pi_n(X, x)$ is abelian, $n \geq 2$.

∘ **Proof** We observe, that when $n \geq 2$, we could obtain two kinds of multiplication with identity e:

...

□

Definition through Loop Space

Factorization through Sphere

1.1.1 Functorial Property

♠ **Proposition 1.1.1.1** The pointed map

$$u : (X, x_0) \to (Y, y_0)$$

induces

$$u_* : \pi_n(X, x_0) \to \pi_n(Y, y_0) \qquad [f] \mapsto [u \circ f]$$

which is a group map.

∘ **Proof** ... □

1.1.2 Coverings

▶ ***Theorem*** (Homotopy Groups and Coverings) Let $p : X \to Y$ be a covering, choose a base point $x \in X$ with $p(x) = y$, then

...

1.1.3 Homotopy Groups of Pairs and Triple,Long Exact Sequence

1.2 n-Connectivity

Definition 1.2.0.1

- A space X is called 0-**connected** if $\pi_0(X)$ consists of one point; (Path-connected)
- A space X is called 1-**connected** if $\pi_k(X) = \bullet, \forall 1 \leq k \leq n$;
- A space X is called ∞-**connected** if $\pi_k(X) = \bullet, k \geq 1$;

And:

Definition 1.2.0.2 Let $f : X \to X$;

- It is called 0-connected if the induced map

$$\pi_0(f) : \pi_0(X, x) \to \pi_0(X, x)$$

is surjective;

- It is called ***n*-connected** of the induced map

$$\pi_k(f) : \pi_k(X, x) \to \pi_k(X, x) \qquad \text{is bijective}, 0 \leq k \leq n-1$$
$$\pi_n(f) : \pi_n(X, x) \to \pi_n(X, x) \qquad \text{is surjective}$$

- It is called **∞-connected** if $\pi_k(f) : \pi_k(X, x) \to \pi_k(X, x)$ is bijective;

♠ **Proposition 1.2.0.3** (Criteria by Sphere) TFAE:

- $\pi_n(X, x)$ is trivial for any base point $x \in X$;
- Every $S^n \to X$ is nullhomotopic. i.e. $[S^n, X] = id_\bullet$;
- Every $S^n \to X$ extends to $D^{n+1} \to X$;

∘ **Proof** ... □

★ **Corollary 1.2.0.4** (Criteria by Sphere) TFAE:

- X is n-connected;
- Given any k with $0 \leq k \leq n$, every map $S^k \to X$ is nullhomotopic;
- Given any k with $0 \leq k \leq n$, every map $S^k \to X$ extends to a map $D^{k+1} \to X$;

∘ **Proof** Omitted. □

Moreover,

★ **Corollary 1.2.0.5** ...

∘ **Proof** ... □

1.3 Colimits

▶ ***Theorem*** (Homotopy Groups and Colimits) Let: X be a topological Hausdorff space, with a sequence of closed subspaces

...

∘ **Proof** ... □

Chapter 2 CW-Theory

2.1 CW-Approxiamation

2.2 Whitehead's Theorem

Chapter 3 Homotopy and Homology

3.1 Hurewicz's Theorem

Chapter 4 Fibration and Cofibration

4.1 Fibrations

4.1.1 Homotopy Lifting Property

4.1.2 Maps into Fibration

4.2 Cofibrations

4.2.1 Homotopy Extension Prorperty

Definition 4.2.1.1 A map $i : A \to X$ has the **homotopy extension property**(HEP) with respect to Y, if every pair of map $g : X \to Y, h : A \times I \to Y$ with

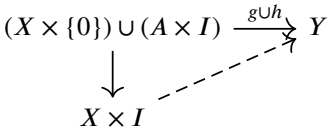
$$h(a, 0) = g(i(a)) \qquad \forall a \in A$$

Admits an extension $H : X \times I \to Y$ satisfying

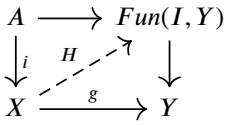
$$H(x, 0) = g(x) \qquad H(i(a), t) = h(a, t)$$

The following diagrams

Map Space:



Maps:



♣ **Example 4.2.1.2** A question is: Is any (closed) inclusion a cofibration? The answer is no. Given:

...

4.2.2 title

Chapter 5 Spectra

Preparations: Recall some operations of spaces

- $\Sigma X := X \wedge S^1$ Reduced suspension;
- $\Omega X := \{\gamma : I \to X | \gamma(0) = \gamma(1)\}$;

5.0.1 Stable Equivalence

5.1 Spectra

5.1.1 Some of Spectrum

Because of adjunction

$$\Sigma \dashv \Omega$$

The spectrum has Σ-Representation and Ω-Representation

Pre-Spectrum

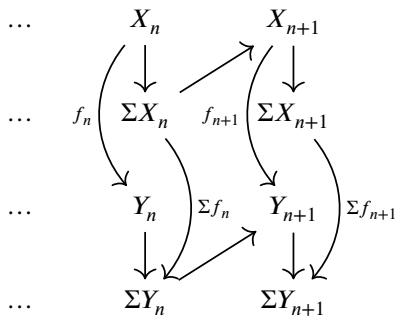
Definition 5.1.1.1 A **prespectrum** or **sequential spectrum** is:

- $E = ((E_n), (\sigma_n : \Sigma E_n \to E_{n+1}))$; or by adjunction
- $E = ((E_n), (\sigma_n : E_n \to \Omega E_{n+1}))$;

The only requirement is: The structure maps are maps between pointed spaces;
Such E_n is called n-level set.

Then, a **morphism** between the is a family of morphisms $(f_n : E_n \to E'_n)$ such that

$$f_{n+1} \circ \sigma_n = \sigma_n \circ (\Sigma f_n)$$



Definition 5.1.1.2 Category of sequential spectrum $\mathcal{S}^{\mathbb{N}}$;

And

Definition 5.1.1.3 Let X be a spectrum, define

...

♣ **Example 5.1.1.4** (Sphere Spectral)

▶ ***Theorem*** ...

Stable Homotopy Groups

In principle, we should not introduce this concept in unstable homotopy theory... But the fact is, we must introduce it here

Definition 5.1.1.5

Spectrum

Definition 5.1.1.6 A **Ω-spectrum** is:

- $((E_n), (\sigma_n : \Sigma E_n \to E_{n+1}))$, and σ_n is a weak equivalence; or
- $((E_n), (\tilde{\sigma}_n : E_n \to \Omega E_{n+1}))$ and $\tilde{\sigma}$ is a weak equivalence;

♣ **Example 5.1.1.7** (Eilenberg-Maclane Spectrum)

5.1.2 Spectrification

♣ **Example 5.1.2.1** (Sphere Spectrum) Consider following spectrum

...

5.1.3 Categoricial Property

5.2 Stability

Chapter 6 Excision Theorem, and Freudenthal Suspension Theorem

Chapter 7 Appendix: Serre's Finiteness Theorem for Spheres

▶ ***Theorem*** (Serre)

$$\pi_k(S^n) = \begin{cases} \mathbb{Z} & n = 0 \\ a \end{cases}$$

7.1 Nilpotence Theorem

7.2 Computing of $\pi_*(S^n)$